$\vec{a}\cdot\vec{b} = |\vec{a}|$ (projection of $\vec{b}$ on $\vec{a}$)

Thus, projection of $\vec{b}$ on $\vec{a} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}|}$.

In the same manner, projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{b}\cdot\vec{a}}{|\vec{b}|}$.

8.8.3 Properties of Scalar Product

(i) Scalar product of two vectors is commutative.

With usual definition, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta = |\vec{b}||\vec{a}|\cos\theta = \vec{b}\cdot\vec{a}$

That is, for any two vectors $\vec{a}$ and $\vec{b}$, $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$.

(ii) Nature of scalar product

We know that $0 \le \theta \le \pi$.

If $\theta = 0$ then $\vec{a}\cdot\vec{b} = ab$ [Two vectors are parallel in the same direction $\Rightarrow$ $\theta = 0$].

If $\theta = \pi$ then $\vec{a}\cdot\vec{b} = -ab$ [Two vectors are parallel in the opposite direction $\Rightarrow \theta = \pi$.].

If $\theta = \dfrac{\pi}{2}$ then $\vec{a}\cdot\vec{b} = 0$ [Two vectors are perpendicular $\Rightarrow \theta = \dfrac{\pi}{2}$].

If $0 < \theta < \dfrac{\pi}{2}$ then $\cos\theta$ is positive and hence $\vec{a}\cdot\vec{b}$ is positive.

If $\dfrac{\pi}{2} < \theta < \pi$ then $\cos\theta$ is negative and hence $\vec{a}\cdot\vec{b}$ is negative.

That is, $\vec{a}\cdot\vec{b}$ is $\begin{cases} > 0 \text{ for } 0 \le \theta < \pi/2 \\ 0 \quad \text{for } \theta = \pi/2 \\ < 0 \text{ for } \pi/2 < \theta \le \pi \end{cases}$

(iii) $\vec{a}\cdot\vec{b} = 0 \Rightarrow |\vec{a}| = 0$ or $|\vec{b}| = 0$ or $\theta = \dfrac{\pi}{2}$

(iv) For any two non-zero vectors $\vec{a}$ and $\vec{b}$, $\vec{a}\cdot\vec{b} = 0 \Leftrightarrow \vec{a}$ is perpendicular to $\vec{b}$.

(v) Different ways of representations of $\vec{a}\cdot\vec{a}$.

$\vec{a}\cdot\vec{a} = |\vec{a}|^2 = (\vec{a})^2 = \vec{a}^2 = a^2$.

These representations are essential while solving problems.

(vi) $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$ and $\hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$ (how?).

(vii) For any two scalars λ and μ

$\lambda\vec{a}\cdot\mu\vec{b} = \lambda\mu(\vec{a}\cdot\vec{b}) = (\lambda\mu\vec{a})\cdot\vec{b} = \vec{a}\cdot(\lambda\mu\vec{b})$.

(viii) Scalar product is distributive over vector addition.

That is, for any three vectors $\vec{a}, \vec{b}, \vec{c}$

$\vec{a}\cdot(\vec{b}+\vec{c}) = \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c}$ (Left distributivity)

$(\vec{a}+\vec{b})\cdot\vec{c} = \vec{a}\cdot\vec{c} + \vec{b}\cdot\vec{c}$ (Right distributivity)

Subsequently,

$\vec{a}\cdot(\vec{b}-\vec{c}) = \vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{c}$ and $(\vec{a}-\vec{b})\cdot\vec{c} = \vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c}$

These can be extended to any number of vectors.

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